

Migrating Pipelines from ADF to Synapse

Table of Contents

1. [Introduction](#)
2. [Problem Statement](#)
3. [Objectives](#)
 - 3.1 [Objective 1: ADF to Synapse Migration](#)
 - 3.2 [Objective 2: Databricks to Fabric Migration](#)
4. [Tools & Technologies](#)
5. [Architecture Diagram](#)
6. [Implementation Details](#)
 - 6.1 [ADF Pipeline Development](#)
 - 6.2 [ADF to Synapse Migration Steps](#)
 - 6.3 [Databricks to Fabric Migration Steps](#)
7. [Deliverables](#)

1. Introduction

This document outlines the project undertaken to migrate existing Azure Data Factory (ADF) pipelines and Databricks notebooks to modern platforms: Azure Synapse and Microsoft Fabric,

respectively. This modernization enhances scalability, maintainability, and integration within Azure's ecosystem.

2. Problem Statement

The current system uses Azure Data Factory for data movement and transformation and Databricks for processing. The aim is to modernize the solution by migrating pipelines, datasets, and notebooks to Azure Synapse and Fabric, maintaining the same functionality while optimizing performance.

3. Objectives

3.1 Objective 1: ADF to Synapse Migration

- Develop ADF pipelines that:
 - Copy multiple tables from Azure SQL to ADLS Gen2.
 - Use Dataflows to clean the data.
- Migrate these pipelines, datasets, and linked services to Azure Synapse Studio in two different ways (manual + ARM Template or JSON export/import).
- Update connections post-migration and validate pipeline functionality.

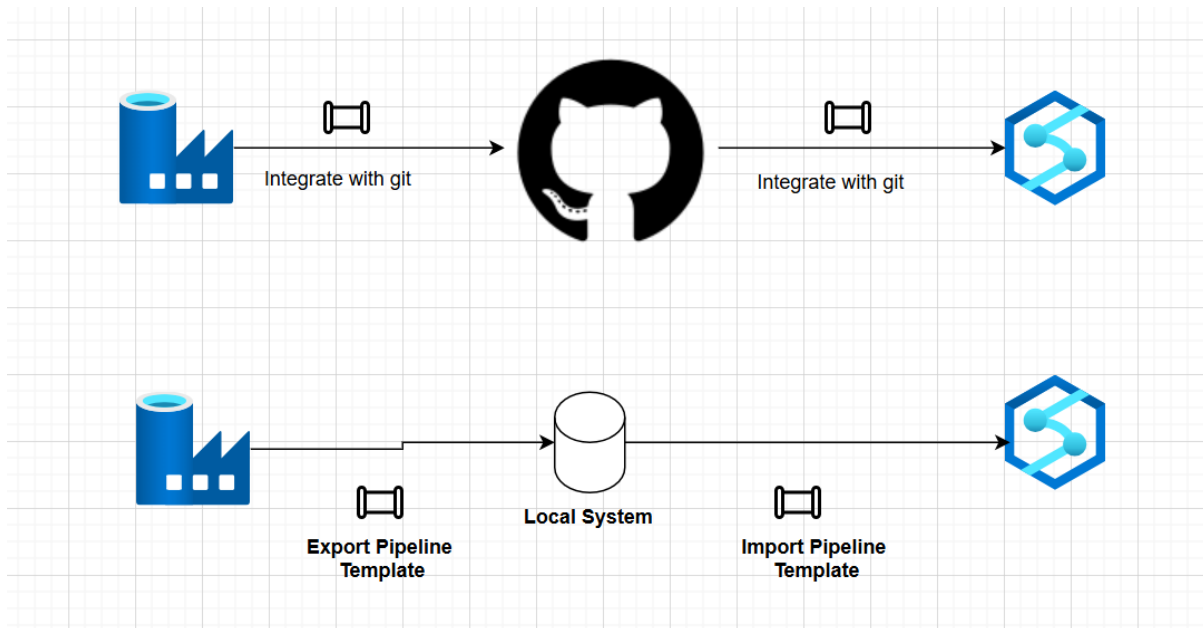
3.2 Objective 2: Databricks to Fabric Migration

- Develop Databricks Notebooks.
 - Migrate these notebooks to Microsoft Fabric.
 - Update mount points/connections as needed.
 - Validate each notebook post-migration.
-

4. Tools & Technologies

- **Azure Data Factory** – Pipeline orchestration
 - **Azure Synapse Analytics** – Unified analytics platform
 - **Azure SQL Database** – Source data
 - **Azure Data Lake Gen2** – Storage layer
 - **Azure Key Vault** – Secrets management
 - **Databricks** – Notebook development
 - **Microsoft Fabric** – Notebook execution post-migration
-

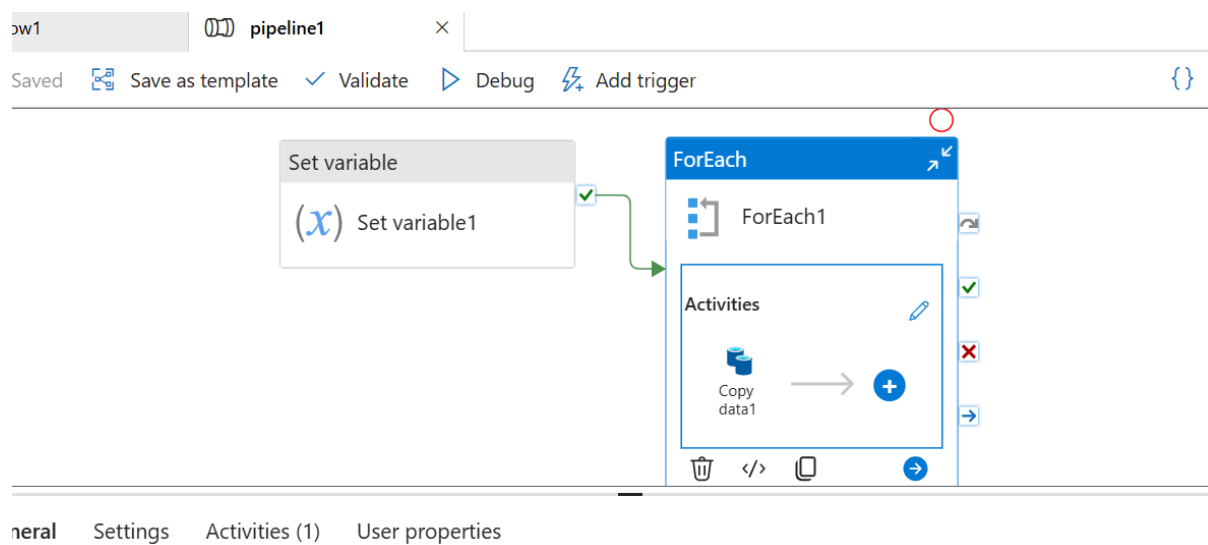
5. Architecture Diagram



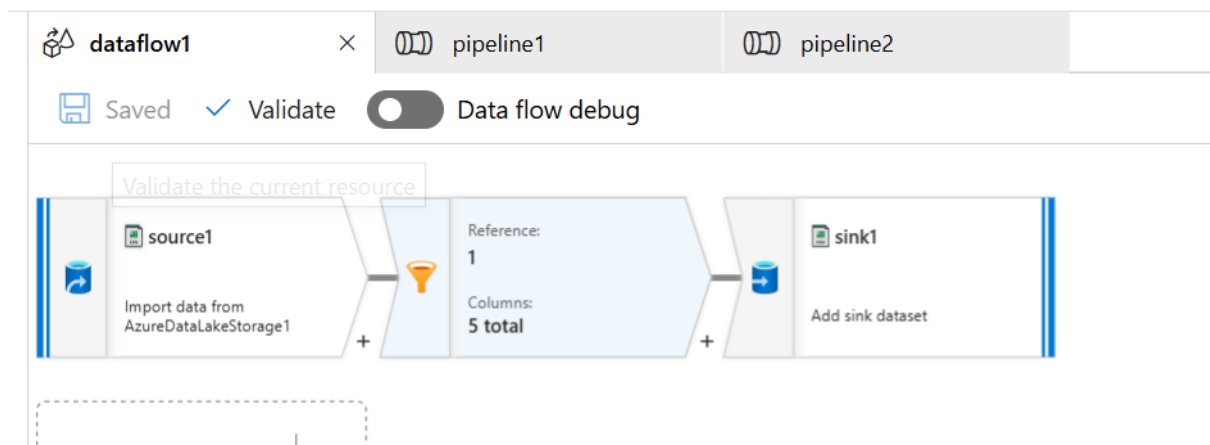
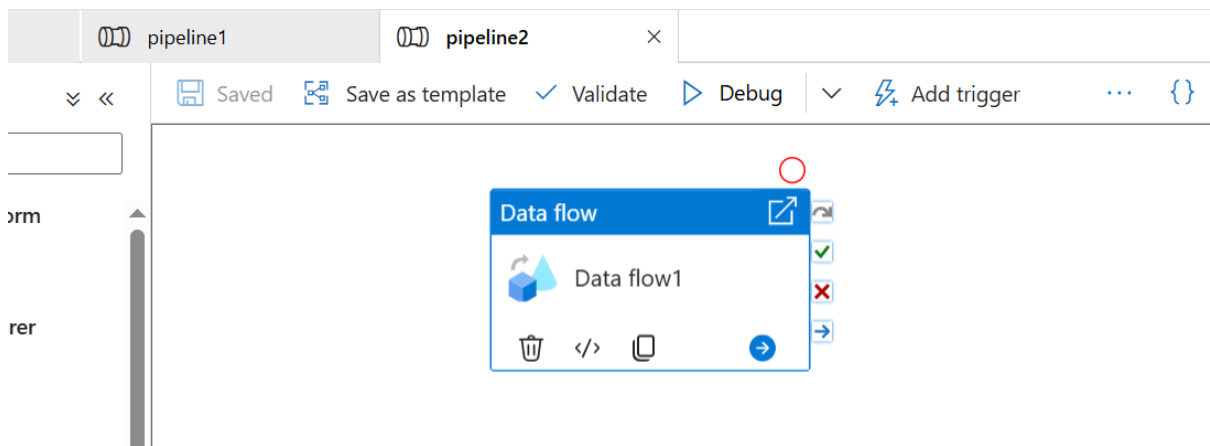
6. Implementation Details

6.1 ADF Pipeline Development

- **Pipeline 1:** CopyMultipleTables using Lookup + ForEach + Copy Activity



- **Pipeline 2:** CleanDataWithDataflows using Mapping Dataflows



- **Linked Services:** Azure SQL, ADLS Gen2, Key Vault
- **Datasets:** Parameterized datasets for reuse

6.2 ADF to Synapse Migration Steps

- **Method 1:** Manual recreation in Synapse Studio using JSON export/import

</

- **Method 2:** ARM Template deployment
- Export pipeline template

> This PC > Downloads > Pipeline 1 > Pipeline 1				Search Pipeline 1	
Name	Type	Compressed size	Passw		
manifest	JSON Source File	11 KB	No		
Pipeline 1	JSON Source File	4 KB	No		

- **Method 3: Import from Github**

Git repository

Git repository information associated with your Synapse workspace. [Learn more](#)

Edit
 Overwrite live mode
 Disconnect
 Import resources

Repository type	GitHub
GitHub account	VishalKanaka
Repository name	ADF_Synapse_Pipeline_Migration
Collaboration branch	main
Publish branch	main
Root folder	/
Last published commit	d054eca946b1a6f9e5eec8c796e43a85b9d11cf7
Custom comment	Enabled

- Successfully imported in synapse

/ vishal branch

Validate all
 Commit all
 Publish

Integrate

Filter resources by name

Pipelines

pipeline1

pipeline2

pipeline2

dataflow1

Committed

Validate

Data flow debug

Reference: 1

Columns: 5 total

filter1

Filtering rows using expressions on columns 'transaction_id, account_id'

sink1

Add sink dataset

Add Source

- **Post-Migration Steps:**
 - Replace linked services with Synapse equivalents
 - Update Key Vault references
 - Run integration tests

6.3 Databricks to Fabric Migration Steps

- Export notebooks from Databricks workspace (.dbc or ipynb)

Workspace > Users > vishal@harpalvaghel02outlook.onmicrosoft.com > AzureDatabricks-To-Fabric-Notebooks-Migration vishal >

Azure ☆

Send feedback

Share

Create ▾

Name ↕	Type	Owner	Created at
Notebook1	Notebook	Vishal	May 11, 2025, 09:4...
Notebook2	Notebook	Vishal	May 11, 2025, 09:3...

AzureDatabricks-To-Fabric-Notebooks-Migration [Send feedback](#)

Branch: vishal ▾

Create Branch

bc76a88 [🔗](#)

Changes Settings

- Import into Microsoft Fabric from github

Connect Git repository and branch

Repository URL

https://github.com/VishalKanaka/AzureDatabricks-To-Fabric-Notebooks-Migration

Git folder

/Azure

Branch

vishal

Disconnect workspace

Vishal-WSP Create deployment pipeline Create app Manage access Wo					
New item New folder Import Migrate Source control Filter by keyword Filter					
	Name	Git status	Type	Task	Owner
	Notebook1	—	Notebook	—	Vishal
	Notebook2	—	Notebook	—	Vishal

- Modify or remove `dbutils.fs.mount()` logic if unsupported

Data items

Resources

Add data items

Items

Lakehouse

Tables

Files

input

input

1

df = spark.read.option("header", "true").csv("Files/input/transactions.csv")

2

[4]

✓ 9 sec - Command executed in 9 sec 804 ms by Vishal on 1:00:23 PM, 5/12/25

PySpark (Python)

Spark jobs (1 of 1 succeeded)

Resources

Log

1

df.show()

[5]

✓ <1 sec - Command executed in 923 ms by Vishal on 1:00:38 PM, 5/12/25

PySpark (Python)

Spark jobs (1 of 1 succeeded)

Resources

Log

transaction_id

account_id

transaction_date

transaction_amount

transaction_type

1

45

2024-01-01

100.50

Deposit

Method 2

- Manual import by uploading the notebooks

Vishal-WSP

Create deployment pipeline

Create app

Manage access

Wo

New item

New folder

Import

Migrate

Source control

Filter by keyword

Notebook

Report, Paginated Report or Workbook

From this computer

Import notebook source code files from your local drive.

Name	Task
Lakehouse	Lakehouse

7. Deliverables

- ☒ Pipelines in ADF and Synapse
- ☒ Four Notebooks in Databricks and Fabric
- ☒ Architecture Diagram (.drawio file)
- ☒ Complete Documentation:
 - Data Ingestion Setup
 - Dataflow Logic
 - Notebook Code
- ☒ Uploaded to GitHub: [<https://github.com/VishalKanaka/AzureDatabricks-To-Fabric-Notebooks-Migration.git>]